DAY 11 | POST-READ

React Basics: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Part A: Summary of today

Topic	What to remember
SPA	Loads once, updates parts of the page, fetches JSON from APIs.
Component	A function returning JSX. Capitalised name. One parent element.
JSX	{expressions}, className, self-closing tags, camelCase events, style objects.
Props	Inputs from the parent, read-only. Destructure: function Card({ title }).
children	Whatever sits between a component's tags; great for layouts.
State	const [value, setValue] = useState(initial). Change only via the setter.
Re-render	Calling a setter re-runs the component with the new value.
Events	onClick, onChange, onSubmit (call e.preventDefault()).
Controlled input	value from state, onChange updates state.
Project setup	npm create vite@latest app -- --template react, npm install, npm run dev.

Part B: Going beyond today

1. Updating state from the previous value

```
setCount(c => c + 1);   // safe even when several updates happen quickly
```

2. State with objects and arrays: copy, do not mutate

```
setUser({ ...user, city: "Osaka" });
setItems([...items, newItem]);
setItems(items.filter(i => i.id !== id));
```

React compares old and new values. If you change the same object in place, React may not notice and the screen will not update.

3. Lifting state up

When two sibling components need the same data, move the state to their closest common parent and pass it down as props, along with functions to change it.

4. Prop types with TypeScript (preview)

```
type CardProps = { name: string; price: number };
function ProductCard({ name, price }: CardProps) { ... }
```

Many teams use TypeScript with React to catch wrong props before the app runs.

5. React DevTools

Install the React Developer Tools browser extension to inspect components, their props and state while the app runs.

Part C: Check yourself

- Why do component names start with a capital letter?
- What is the difference between props and state?
- Why should you never write count = count + 1 for state?
- What does a controlled input mean?
- Why is setItems([...items, x]) better than items.push(x)?

Part D: Practice ideas for tonight

- Build a ProfileCard component and render three cards with different props.
- Build a like button that toggles between liked and not liked and counts total likes.
- Build a temperature converter with a controlled input that shows Fahrenheit as you type Celsius.

Coming up on Day 12: conditional rendering, rendering lists with keys, fetching data with useEffect, and connecting React to a FastAPI back end.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today